



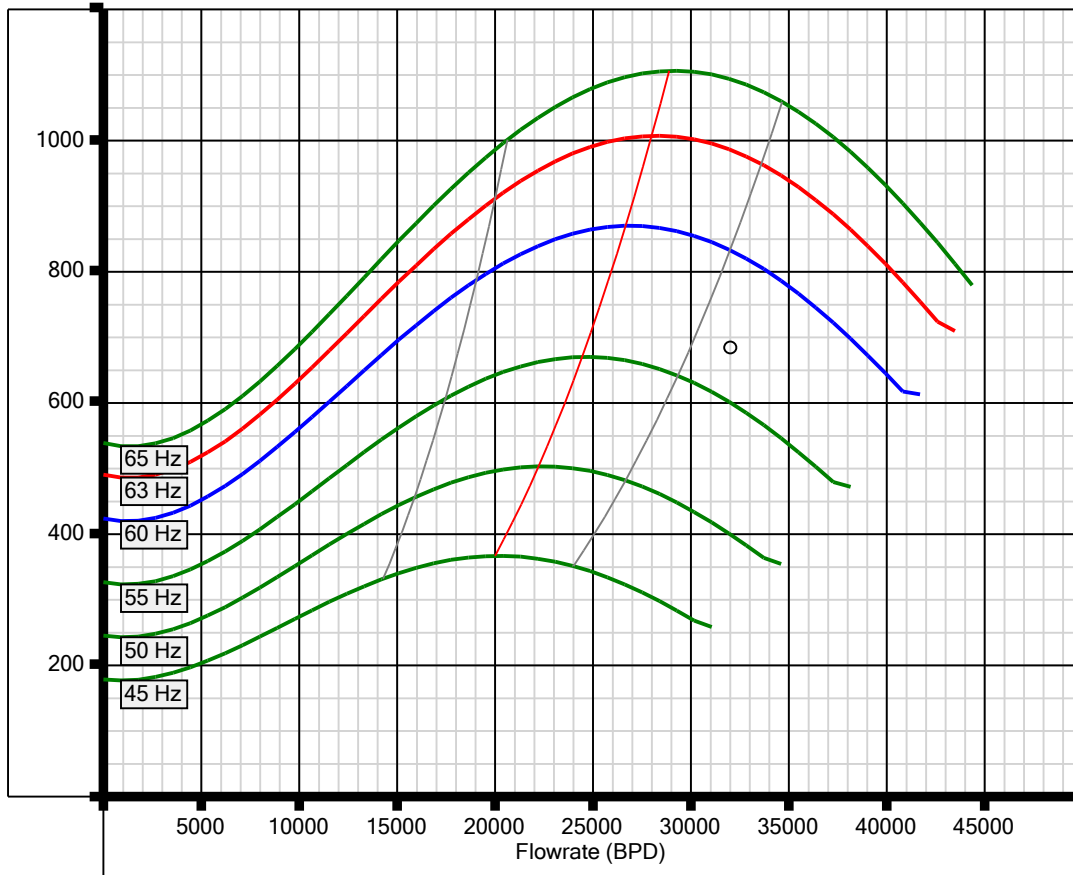
Project: AutographPC example for gassy wells
Customer: Sertecpet
Well: HPS
Engineer: Carlos Villacres
Country: EC
State:
County:
Well API#:
Reservoir Name:

Pump: 39-675HC27000Standard
ThChmbr: HDTC
Motor: Teco TEFC 800 HP 460 V/ 867 A (60 Hz)
Controller: VSD 8700-VT 748kVA 480V/ 900A
Sensor: none

Comments:

39-675HC27000Standard

Power (HP)





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Comments:

Operating Parameters / Selection:

Design Point:

Desired flow (total) 32000 BPD
Frequency 56.9 Hz
Boost Pressure 600.0 psi

Pump Selection:

	<u>Intake</u>	<u>Discharge</u>
Pressure	120.0 psi	720.0 psi
Flowrate	32000 BPD	32000 BPD
Specific Gravity	0.92 rel-H2O	0.92 rel-H2O
Viscosity	50.0Cp	50.0Cp

Pump Selected:

39-675HC27000Standard
Pump eff = 47.72%
Pshaft RPM = 3395
Shft HP @ 56.9 Hz = 684.7
Shaft load =(Std 88%) / (HS 56%)
Required Motor HP at 60.0 Hz = 729.6

Thrust Chamber Selection:

Type HDTC
No sand present
Pump uses fixed-type stages
Thrust Load 27.6 %
ShaftHPcap 1160 HP

Pressure Data:

Max Working Pressure 1323 psi
Max Allowable Working Pressure 1700 psi

Surface Motor Selection

Motor Selected	=Teco TEFC 800 HP 460 V 867 A	Max Elevation	= 3300ft
Terminal Voltage	=435.5 V	Max Amb Temp	=104.0°F
Motor Current	=792.7 A	Power Type	=UNFILT
Load acc to N.P.	=91.2 %	Frame	=6808A

No comments

Controller Selection:

Surface KW	543.9kW	Voltage Input	480.0V
Input kVA	679.8kVA	Max Well Head Volts	435.5V
System kW	555.0kW	Target Frequency	56.8Hz (8.45V/Hz)
Max Ctrl Current	792.7A	Step-up Trafo	1.0 ratio
Power Cost/kWH	0.05\$/kWH	Selected:	VSD 8700-VT 748kVA/ 480V/ 900A
Total Power Cost	= \$19981/month		

No comments